

System Parameters (defined by hardware)		Profiles (usage of each component mode - defined by software and usage)		
form the datasheets		"off"	"sensing"	"interactive"
Processor				
Active	76 mW	0%	1%	20%
Idle	53 mW	0%	4%	10%
Deep Sleep	0 mW	100%	95%	70%
LED				
On	0 mW	0%	0%	0%
Sensor				
On	13 mW	0%	0%	100%
Idle	2 mW	0%	100%	0%
Off	0 mW	100%	0%	0%
Display				
On	0 mW	0%	0%	0%
Off (leakage)	0 mW	0%	0%	0%
Radio				
Data Rate	13,000 bps	0%	0%	0%
Standby Power	1 mW	0%	100%	95%
TX Power	65 mW	0%	0%	3%
RX Power	40 mW	0%	0%	2%
Battery				
Capacity	250 mAh			
Nominal Voltage	4 V			
Regulator Efficiency	93%			

REFLECTIONS : WHAT DID YOU LEARN FROM ANALYZING YOUR POWER. TALK ABOUT SOME POTENTIAL TRADEOFFS.

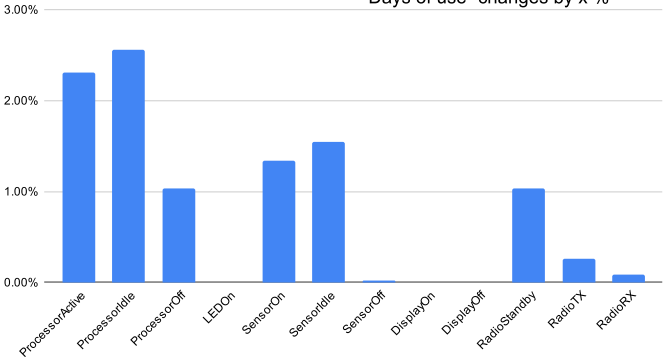
Based on the power analysis, I determined the days of use metric by modeling a typical day across three usage profiles: off, sensing, and interactive, and weighting each by the expected hours per day. This approach captures realistic behavior rather than relying on worst case or peak power scenarios. Using this model, a 210 mAh battery provides approximately 7.4 days of operation, while even a much smaller 50 mAh battery still supports about 1.7 days of typical use.

Given these results, an optimal battery size for the wristband is likely in the 200 to 300 mAh range. This size provides ample margin for variability in usage, battery aging, and inefficiencies, while remaining small and comfortable for a wearable form factor. The analysis also shows that the system could function for a full day with a significantly smaller battery if needed.

Several tradeoffs could further improve the user experience. From a software perspective, minimizing radio usage by computing velocity on-device and transmitting only compact result packets keeps power consumption low without reducing functionality. From a hardware perspective, aggressively using sensor low power modes and processor sleep states improves battery life at the cost of slightly increased latency and implementation effort. Increasing responsiveness or streaming more data would improve immediacy and fidelity but would require higher power draw, a larger battery, or additional development complexity.

Sensitivity Analysis

If component changes by 10%...
“Days of use” changes by x %



Total power in profile (mw)

Maximum Time

"off"

0.4455 mW

1931.0 hours

"sensing"

5.99155 mW

143.6 hours

"interactive"

37.3303 mW

23.0 hours

Effective Battery Capacity

860.25 mW*h

Days of Use

8.82 days

Hours of Use

211.77 hours